

LLMOrchestrator cross-org mirror divergence — reconciliation plan

Date: 2026-06-08 **Scope:** submodules/llm_orchestrator gitdir only. READ-ONLY investigation. No push / merge / force performed. **Investigator note:** All facts below are from actual git output, not surmise (§11.4.6).

1. Remote map

`git -C submodules/llm_orchestrator remote -v:`

Remote	Fetch URL	Push URL
origin	git@github.com:vasic-digital/LLMOrchestrator.git	git@github.com:HelixDevelopment/LLMOrchestrator.git
gitlab	git@gitlab.com:vasic-digital/LLMOrchestrator.git	git@gitlab.com:vasic-digital/LLMOrchestrator.git
github	git@github.com:HelixDevelopment/LLMOrchestrator.git	git@github.com:HelixDevelopment/LLMOrchestrator.git
upstream	git@github.com:HelixDevelopment/LLMOrchestrator.git	git@github.com:HelixDevelopment/LLMOrchestrator.git

Interpretation:

- **§6.W pin targets (what Lava consumes):** the vasic-digital pair — origin *fetch* (github.com/vasic-digital) + gitlab (gitlab.com/vasic-digital). The local master tracks [origin/master].
- **Cross-org HelixDevelopment mirror:** github and upstream are BOTH github.com/HelixDevelopment/LLMOrchestrator.git (duplicate remotes pointing at the same URL).
- **MISCONFIGURATION (forensic finding):** origin has a **split URL** — fetch = vasic-digital, push = HelixDevelopment. This is almost certainly the cause of the original session's confusion: a `git push origin` would have pushed to HelixDevelopment (the diverged side) while `git fetch/tracking` reads from vasic-digital. This split is itself worth flagging to the operator (see §6).

Local branch state: * master a484f7d [origin/master: ahead 1] — clean working tree (no uncommitted changes).

2. Divergence facts (from `git rev-list --left-right --count + git merge-base`)

Fact	Value
Local master (vasic-digital pin)	a484f7dd
upstream/master (HelixDevelopment)	80ead875968906bfe71f874d2de51cd7b6c760d7
Commits local has, upstream lacks	1 (a484f7d)
Commits upstream has, local lacks	1 (80ead87)
Common ancestor (merge-base)	d2a2151 (cascade: chore(1.1.8-dev-rc1 cascade)...)
Relationship	True divergence — NOT a fast-forward in either direction
Histories	Related (shared ancestor exists — NOT unrelated histories)
upstream/main (separate branch)	bde3643 — not involved in this reconciliation

Conclusion: this is a clean 1-vs-1 fork off a shared base. Neither side is a superset of the other; each carries one independent commit.

3. What each side contains

Local a484f7d (vasic-digital side — the Lava pin)

```
chore(deps): add $11.4.31 helix-deps.yaml dependency manifest
helix-deps.yaml | 32 +++++ (1 file, +32)
```

Adds the helix-deps.yaml manifest (the work from this session). helix-deps.yaml is **ABSENT** on the HelixDevelopment side.

Upstream 80ead87 (HelixDevelopment side)

chore(go.mod): remove dead llmprovider replace + stray analysis artifact (Wave4 W4E)

```
go.mod | 2 --
submodule-analysis.txt | 8 ----- (2 files, -10)
```

Dated 2026-06-04 (4 days older than the local commit). Removes a **dead replace digital.vasic.llmprovider => ../LLMProvider** directive from go.mod:18 and deletes a stray submodule-analysis.txt. Commit body asserts go build/go vet/go mod verify all green.

Verified on the local (vasic-digital) side, this cleanup is still UN-applied:

- go.mod:18 on local master **still contains** replace digital.vasic.llmprovider => ../LLMProvider.
- submodule-analysis.txt is **still PRESENT** on local master.

So the HelixDevelopment commit is **genuine, useful, not-yet-superseded cleanup work** — not a stale fork. It should be preserved, not orphaned.

Verdict

Neither side is stale or a superset. The two commits are **complementary** (disjoint files, both wanted). The correct outcome is a UNION of both.

4. Recommended reconciliation — OPTION (a): merge HelixDevelopment into local, then push to all three

Why (a) and not (b)/(c):

- **(b) -s ours** would orphan 80ead87's real go.mod/cleanup work — wrong, because that cleanup is genuine and still un-applied locally. Rejected.
- **(c) treat HelixDevelopment as an independent fork** — abandons a legitimate cleanup commit and leaves the vasic-digital pin carrying a dead replace directive the upstream already fixed. Rejected.
- **(a) merge** — preserves BOTH commits, is **conflict-free** (verified), and is the §6.T.3-compliant non-force path. The -s ours 4-mirror reconciliation pattern is reserved for "sibling mirror has diverged AND its content is superseded"; here the content is NOT superseded.

Conflict-free confirmation (read-only): `git merge-tree --write-tree master upstream/master` exited 0 and produced a clean tree (9fed01f...) with **no conflict markers**. The merge touches disjoint files (`helix-deps.yaml` added by us vs `go.mod/submodule-analysis.txt` cleaned by them), so it merges cleanly.

Exact commands (to run AFTER operator approval — see §5)

```
cd submodules/llm_orchestrator
```

```
# 0. Safety: confirm clean tree + current SHAs
```

```
git status --porcelain          # expect empty
```

```
git rev-parse master            # expect a484f7dd...
```

```
git rev-parse upstream/master   # expect 80ead875...
```

```
# 1. Merge the HelixDevelopment commit into local master (non-FF, no force)
```

```
git merge --no-ff upstream/master \  
-m "merge: reconcile HelixDevelopment cross-org mirror (80ead87  
go.mod cleanup) into vasic-digital master"
```

```
# expected result: clean merge tree 9fed01f..., no conflicts
```

```
# 2. Verify build still green after the merge (the upstream commit claims green)
```

```
go build ./... && go vet ./... && go mod verify
```

```
# 3. Push the merge to ALL THREE mirror endpoints so they converge on the SAME SHA
```

```
git push gitlab master          # gitlab.com/vasic-digital
```

```
git push origin master          # NOTE: origin PUSH url =  
HelixDevelopment (split-URL!) – see §6
```

```
git push github master          # github.com/HelixDevelopment  
(explicit)
```

```
# For the vasic-digital GitHub endpoint, origin's FETCH url is vasic-digital but its
```

```
# PUSH url is HelixDevelopment. To push the vasic-digital GitHub mirror explicitly,
```

```
# the operator must either fix origin's push URL or add a dedicated remote first
```

```
# (see §6 – this is an operator decision, not a blind push).
```

```
# 4. Confirm convergence (§6.C 2-mirror / cross-org parity)
```

```
git ls-remote gitlab refs/heads/master
```

```
git ls-remote github refs/heads/master
```

```
git ls-remote origin refs/heads/master # fetch url = vasic-  
digital github
```

```
# all three SHOULD report the same merge SHA after the pushes
```

Note on the parent index: the merge advances the submodule HEAD from a484f7dd to the new merge SHA. Per this task's constraints I did NOT touch the parent index. If/when the operator approves the merge+push, whether to re-pin the Lava parent to the merge SHA is a SEPARATE decision (the Q9

always-track-upstream waiver covers HelixQA only, not LLMOrchestrator — LLMOrchestrator is a frozen-by-default pin, so a parent re-pin is a deliberate operator action).

5. OPERATOR-APPROVAL GATE (explicit)

The following require explicit operator authorization BEFORE execution — none were done in this read-only pass:

1. **The merge itself (`git merge --no-ff upstream/master`).** It alters local master history (advances HEAD to a merge commit). Compatible + conflict-free, but still a history-affecting action on a branch that feeds the Lava pin.
2. **Every push (`git push gitlab/origin/github master`).** Per §6.T.3, no push without explicit per-operation approval. Authorization for one push does not extend to the others.
3. **The origin split-URL situation.** `origin fetch=vasic-digital, push=HelixDevelopment`. The operator must decide how to push the **vasic-digital GitHub** mirror — fix origin's push URL back to vasic-digital, or add a dedicated `github-vd` remote. Pushing blindly to origin would land on HelixDevelopment, not vasic-digital. This is a configuration decision, not a mechanical step.
4. **Re-pinning the Lava parent** to the merge SHA (frozen-by-default pin; deliberate operator action; NOT covered by the HelixQA Q9 waiver).

No `-s ours` is recommended, so there is no orphan-commit scenario to approve. If the operator instead preferred Option (b)/(c) (NOT recommended), that WOULD orphan `80ead87` and would need explicit approval to discard that cleanup work.

6. Additional forensic flag for the operator

The origin remote has a **split fetch/push URL** (`fetch = github.com/vasic-digital, push = github.com/HelixDevelopment`). Combined with duplicate `github/upstream` remotes both pointing at HelixDevelopment, this remote layout is the most likely root cause of the original "push to vasic-digital succeeded but HelixDevelopment rejected" confusion this session. Recommend the operator normalize the remote set (one clear remote per mirror endpoint) as a follow-up so future pushes are unambiguous.